AI CUSTOMER LIFE CYCLE VALUE AND PERSONALIZATION WITH GENAI ADVISOR

1. Problem Statement

Banks and payment platforms manage large volumes of customer transaction data but often lack intelligent systems that can accurately estimate the long-term value of each customer. Without this insight, it becomes difficult to prioritize high-value customers, identify at-risk segments, and design personalized retention or cross-selling strategies.

Traditional analytics methods provide only static summaries and are often not sufficient for understanding future customer worth or explaining why a customer belongs to a certain segment. Additionally, business users may find raw model outputs difficult to interpret without contextual insights.

Therefore, there is a need for an end-to-end intelligent system that can:

- predict **Customer Lifetime Value (CLV)**,

- segment customers based on behaviour and value,

- and provide **explainable, business-friendly insights** through an interactive interface.

2. Proposed Solution

To address this problem, we developed an **AI-driven Customer Lifetime Value (CLV) Prediction and Explainability System** that combines machine learning, customer segmentation, and retrieval-augmented generation (RAG).

The system uses a **synthetic customer dataset** to simulate realistic customer transaction and behavioural patterns. From this data, **RFM (Recency, Frequency, Monetary)** and additional behavioural features are engineered to train a regression model for CLV prediction. Multiple models were evaluated, including **Linear Regression**, **Random Forest**, and **XGBoost**, with **XGBoost** selected as the final model due to its superior predictive performance.

To enhance business usability, the system also performs:

- **K-Means clustering** to identify behaviour-based customer groups,
- **Decision Tree-based segmentation logic** to create interpretable value categories (Low / Medium / High),

- and a **RAG-based chatbot** that explains predictions using business rules and dataset-driven insights.

The entire system is integrated into a **Streamlit web application** that supports:

-single customer prediction,

-batch CSV upload,

- data validation,

-optional data cleaning,

-prediction downloads,

-interactive explainability.

3. Objectives

The key objectives of this project are:

- To build a machine learning system for **predicting customer lifetime value**
- To segment customers into **business-friendly value groups**
- To identify **behavioural customer clusters** using unsupervised learning
- To provide **explainable and contextual AI-generated responses**
- To create an easy-to-use **Streamlit-based interface** for both single and batch predictions
- To support business decision-making for **retention, personalization, and customer prioritization**

4. Dataset Overview

The system was developed using a **synthetic customer dataset** containing **100,000 customer-level records**.

Each record represents a single customer and includes transactional and behavioural attributes such as:

- purchase frequency,
- total spend,

- average order value,
- recency,
- customer tenure,
- and engagement-related indicators.

Feature Engineering

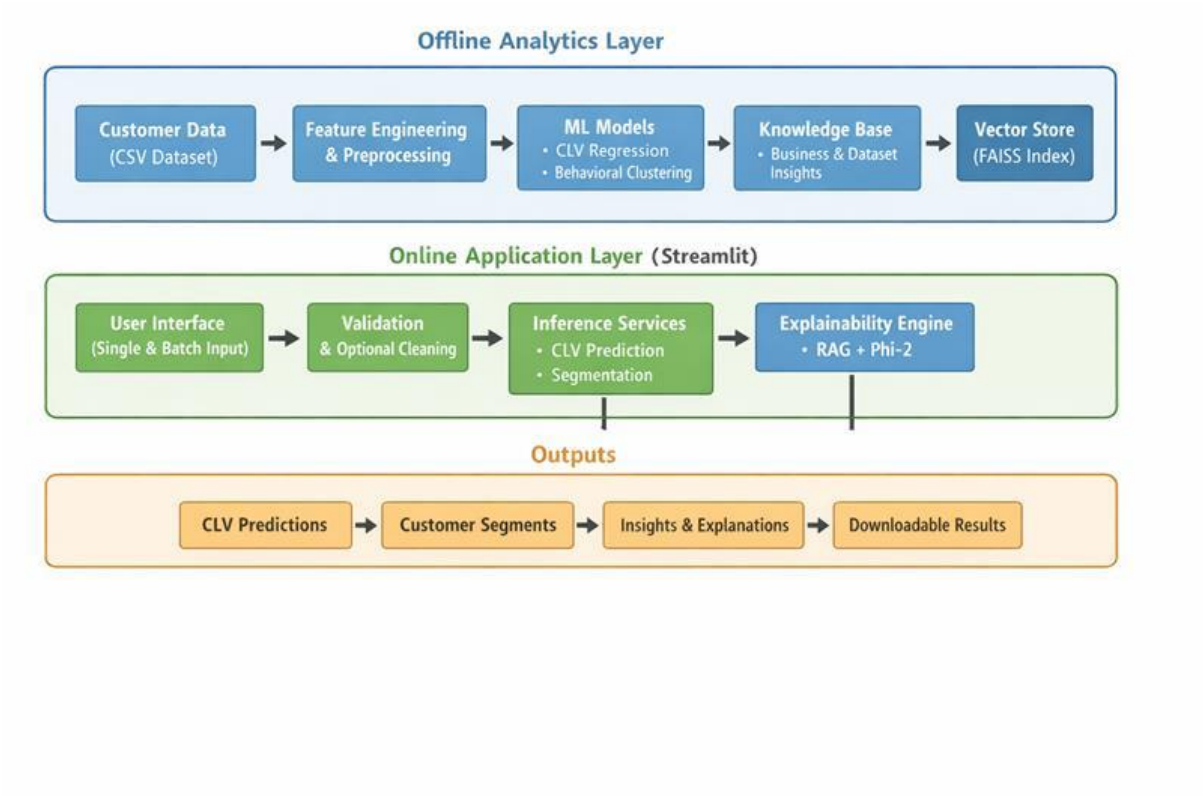
From the raw dataset, the following features were engineered:

- **RFM Features**
    - Recency
    - Frequency
    - Monetary Value
- **Behavioural Features**
    - Average Order Value
    - Customer Tenure
    - Repeat Purchase Tendency
    - Purchase Consistency
    - Engagement-related indicators

Target Variable

- **Customer Lifetime Value (CLV)**

6. System Architecture and Key Components



1. Customer CSV Dataset (100K Records)

What it does:
Serves as the base input data for training and experimentation

2. Data Preprocessing

What it does:
Cleans and standardizes the dataset before model training by handling:

- duplicates,
- missing values,
- formatting inconsistencies.

Why it is used:
Ensures higher data quality and improves the reliability of downstream models.

3. Feature Engineering

What it does:
Transforms raw customer records into model-ready features such as:

- RFM metrics,

- average order value,
- tenure,
- behavioural indicators.

Why it is used:
Raw transactional data alone is often not sufficient for accurate CLV estimation. Engineered features capture patterns that better reflect customer value and purchasing behavior.

4. CLV Model Training

What it does:
Trains regression models to estimate future customer lifetime value.

Models evaluated:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor

Final model selected:

- **XGBoost**

Why it is used:
XGBoost provided the best predictive performance among the evaluated models and is well-suited for structured tabular data with nonlinear feature relationships.

5. Behavioural Clustering (K-Means)

What it does:
Groups customers into clusters based on behavioural similarities.

Why it is used:
Helps identify natural customer segments beyond simple value-based thresholds, enabling richer business understanding of purchasing patterns and customer types.

6. Decision Tree for Segmentation

What it does:
Translates clustering and/or customer features into interpretable segment logic, typically mapping customers into:

- Low Value
- Medium Value
- High Value

Why it is used:
Decision Trees provide transparent rules that are easy for business users to understand and explain.

7. Knowledge Chunking

What it does:
Breaks business rules and dataset insights into structured text chunks for retrieval.

Why it is used:
Smaller, semantically meaningful chunks improve the quality of retrieval in the RAG pipeline.

8. Text Embedding

What it does:
Converts knowledge chunks into vector representations using embeddings.

Why it is used:
Embeddings enable semantic search, allowing the system to retrieve relevant context based on meaning rather than exact keyword matching.

9. FAISS Vector Index

What it does:
Stores embedded knowledge chunks for efficient similarity search.

Why it is used:
FAISS enables fast retrieval of relevant context during real-time chatbot queries, making the explainability layer scalable and responsive.

10. Streamlit User Interface

What it does:
Provides an interactive web interface for:

- single customer input via sliders/forms,
- batch CSV upload,

- viewing predictions and insights.

Why it is used:
Streamlit allows rapid development of a clean, lightweight, and interactive ML application without requiring full-stack web development.

8. End-to-End Workflow

The end-to-end system flow is as follows:

1. A user enters customer details manually or uploads a CSV file.
2. The system validates the input data.
3. Optional preprocessing / cleaning is applied.
4. The engineered feature pipeline prepares the input in the required format.
5. The trained XGBoost model predicts CLV.
6. The customer is assigned to a value segment.
7. Behavioral cluster mapping provides additional context.
8. If the user asks a question, the RAG chatbot retrieves relevant knowledge from FAISS.
9. Phi-2 generates a contextual, explainable response.
10. Final outputs are displayed and can be downloaded.

5. Use Case

This system can be used by:

- **Banks** to identify high-value customers and reduce churn
- **Fintech platforms** to personalize offers and campaigns
- **Marketing teams** to target customers based on predicted future value
- **Business analysts** to understand customer segments with explainable AI

6. Constraints / Limitations

- The training dataset is **synthetic (generated)**, so real-world business behaviour may not be fully captured
- Model performance may vary when deployed on real customer data

- Explainability depends on the quality of knowledge chunks stored in the FAISS index
- Batch prediction quality depends on correct schema and clean input data
- Current system uses predefined segmentation logic and may require retraining for new business contexts

7. Future Scope

- Train and validate on **real-world banking / fintech customer data**
- Improve CLV prediction with **time-series or deep learning models**
- Add **real-time offer recommendation engine** based on predicted CLV and segment
- Expand chatbot knowledge base with **dynamic business rules and policy documents**
- Deploy as a scalable **cloud-based enterprise application** with API integration

Requirements Traceability Matrix (RTM)

Requirement ID	Requirement Name	Implementation Description	Status	If Partial / Skipped – Explanation
FR1	Dataset Upload	CSV upload implemented using Streamlit file uploader. Schema supports CustomerID, Transactions, AvgSpend, Frequency, Tenure, Channel.	Done	
FR2	Data Processing	Data cleaning and preprocessing implemented using Pandas (duplicate handling, missing value checks, normalization, encoding).	Done	
FR3	Feature Engineering	Behavioral features	Done	

		engineered including RFM metrics, tenure, transaction frequency, and spending aggregates.		
FR4	Regression Model (CLV)	CLV regression models implemented including Linear Regression, Random Forest, and XGBoost (offline training).	Done	
FR5	Customer Segmentation	Behavioral segmentation using K-Means clustering. Customers categorized into Low, Medium, and High value segments.	Done	
FR6	Model Comparison	Model evaluation performed using RMSE, R^2 for regression and Silhouette Score for clustering.	Done	
FR7	Personalization Engine	Rule-based mapping between customer segment and offer type designed conceptually.	Partial	Automated recommendation engine not fully implemented in the UI.
FR8	Business Insights	Business insights derived from customer behavior (e.g. high-frequency and high-spend customers drive majority revenue).	Done	

FR9	GenAI RAG Chatbot Objective	RAG chatbot implemented using Phi-2 for embeddings and generation with FAISS for semantic retrieval.	Partial	Performance limited due to local Phi-2 latency and context window size.
FR10	GenAI Supported Questions	Chatbot supports natural language queries related to top customers, profitability, and retention.	Done	
FR11	GenAI Data Sources	RAG system grounded on model outputs, segmentation results, and business insight chunks.	Done	
FR12	Explainable Output	Natural language explanations generated describing high-value customers and suggested targeting strategies.	Done	